

Transmon-Resonator System: Parameter Identification

α (anharmonicity)

$\approx -E_C$ from Q3D

f_{qubit} (qubit frequency)

$\approx \sqrt{(8 E_J E_C)} - E_C$ from Q3D

f_{res} (resonator freq.)

HFSS eigenmode solver

κ (resonator linewidth)

HFSS via 50Ω port loss